

Reproducing and Interrogating a Multi-Omics Survival Foundation Model Across the TCGA Pan-Cancer Atlas

A faithful, HPC-scale re-implementation of the SeNMo preprocessing pipeline — 33 cohorts, 6 omics, 81,240 features — with an exploratory and representational analysis of its learned embeddings.

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01 Background & Objective

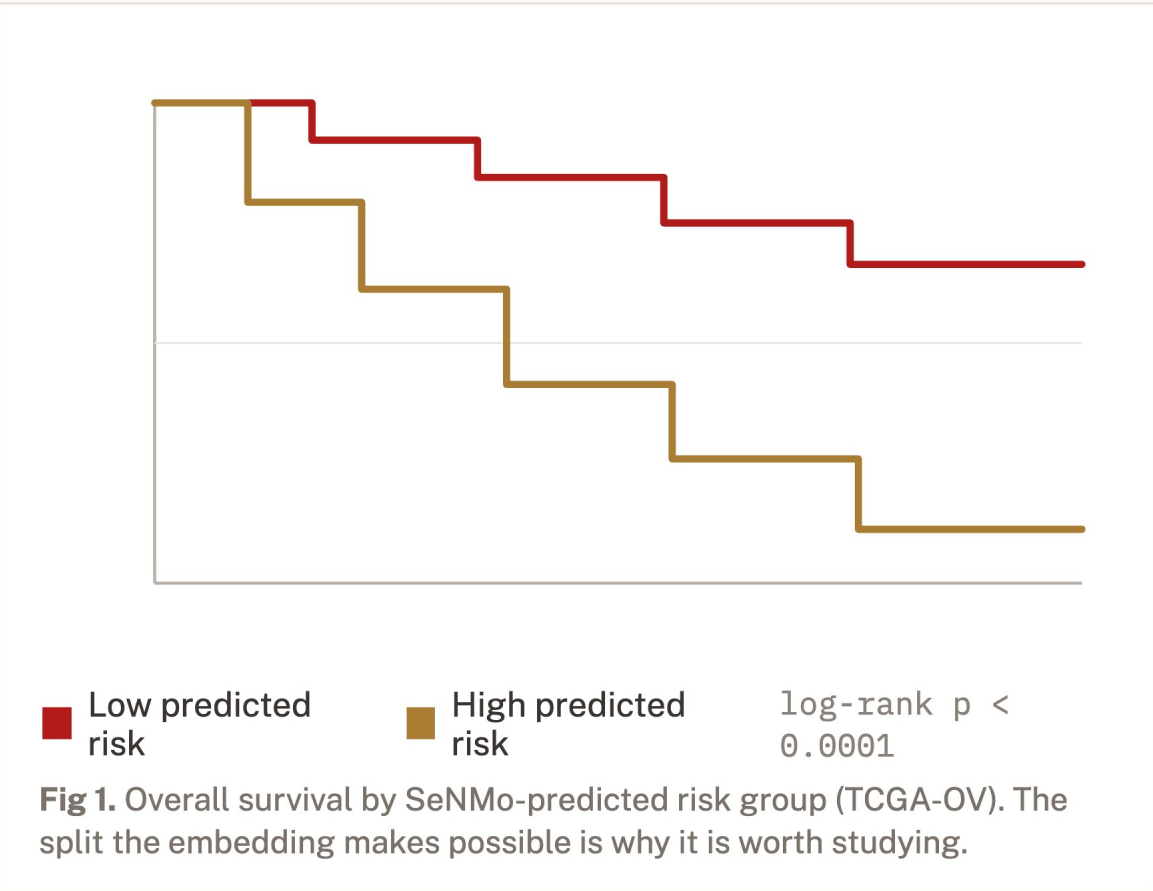
SeNMo is a self-normalizing multi-omics foundation model that predicts cancer-patient survival (Cox objective; concordance index) by compressing roughly **80,697** molecular and clinical measurements per patient into a **48-dimensional** embedding across 33 TCGA cancers. Faithful use of its published checkpoints requires reproducing the exact feature matrix it was trained on.

OBJECTIVE

Reproduce the authors' six-omic preprocessing end-to-end at pan-cancer scale, verify it against their published dimensionalities, and interrogate what the learned embedding actually encodes.

02 Cohort & Exploratory Analysis

Every cohort was characterized from first principles — Kaplan-Meier survival, Cox regression, and the concordance index — before any model saw the data. In the TCGA-OV cohort (428 women, high-grade serous), patients stratified by SeNMo's predicted risk separate sharply.



Clinical-only Cox baseline **C-index 0.626**. mRNA↔protein agreement varies widely (PTEN $r = 0.75$, BRCA2 $r = 0.10$) — the empirical case for reading all six omics together.

03 The Model



04 Methods

Six omics are preprocessed per cohort with the authors' filters — variance threshold, drop-constant, drop-duplicate, correlation pruning — then merged on a shared sample key with survival labels held in raw units.



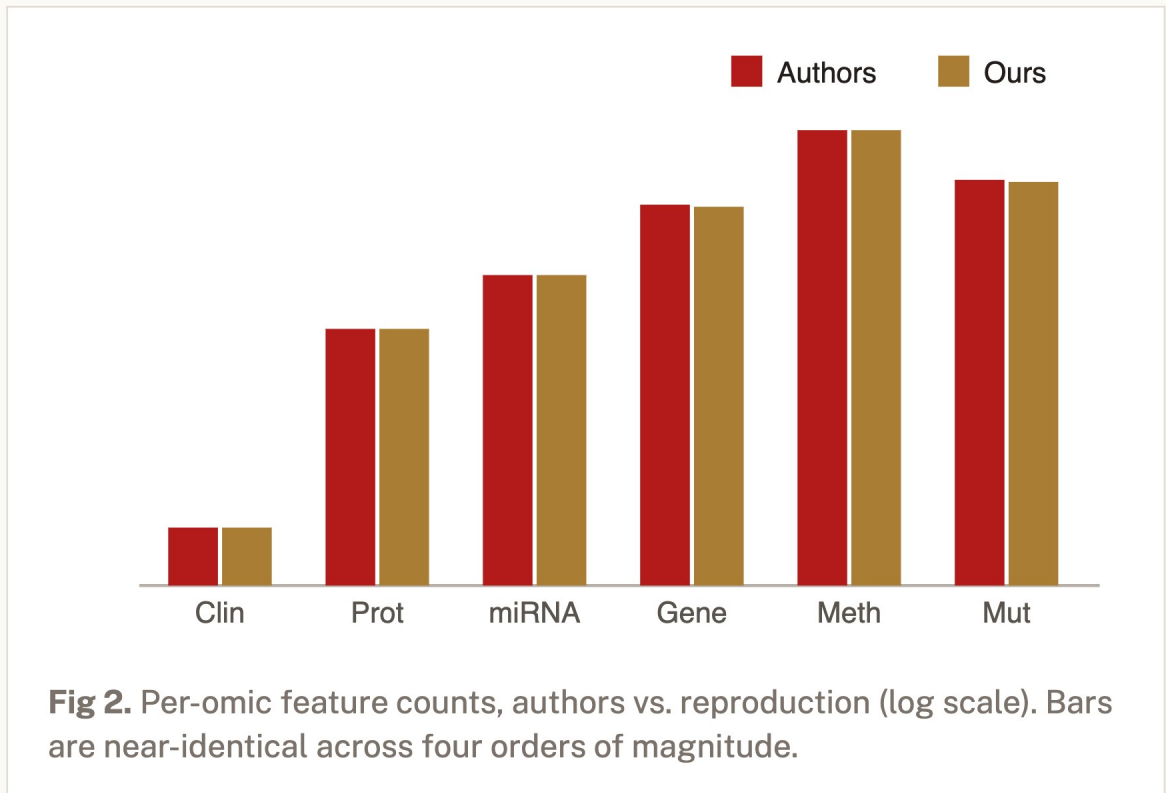
At scale: Slurm job arrays processed all 33 cohorts in parallel on the Cornell Cayuga HPC; **Parquet** checkpoints replaced wide CSV, eliminating out-of-memory failures on ~130,000-column matrices.

05 Reproduction Results

The assembled matrix is **11,127 patients × 81,240 features** — within **0.7%** of the authors' 80,697. Two omics match to the exact feature.

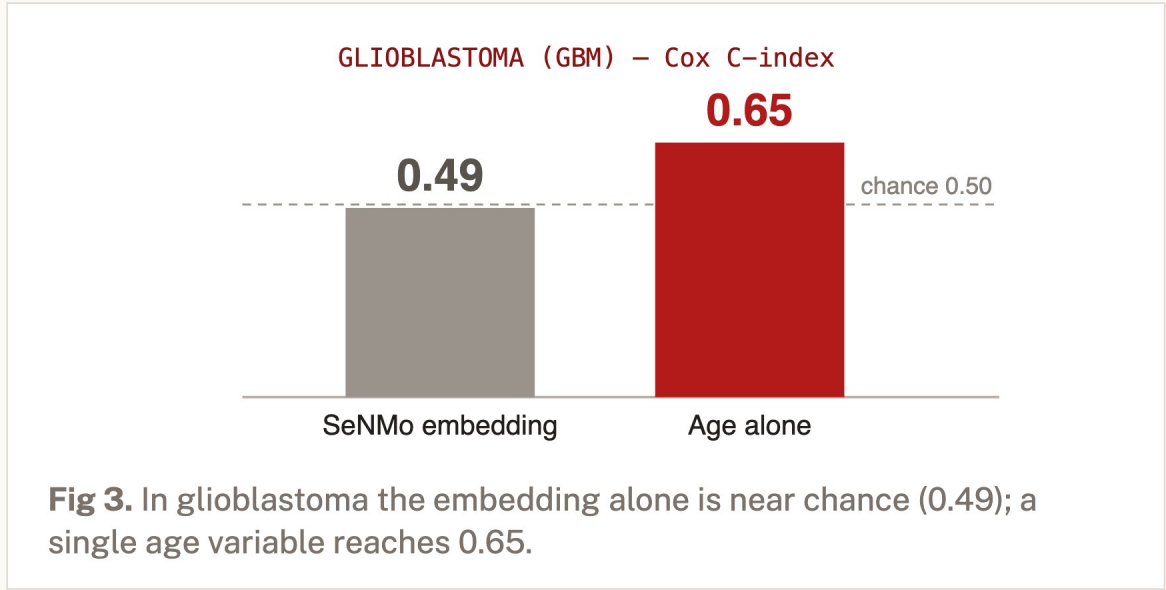
OMIC	AUTHORS	OURS	Δ%
Clinical	4	4	0.0
Protein (RPPA)	472	472	0.0
miRNA	1,730	1,746	+0.9
Gene expression	8,794	8,582	-2.4
DNA methylation	52,396	53,387	+1.9
DNA mutation	17,301	17,078	-1.3
6-MODAL TOTAL	80,697	81,240	+0.7

06 Feature-Count Fidelity



07 A Representational Gap

Probing the frozen 48-dim embedding showed the 80k→48 bottleneck **systematically discards patient age** (mean $r \approx 0.05$ across 23 cancers), yet re-adding it improves prediction — most where molecular signal is weakest.



08 Conclusions

- Full six-omic pipeline reproduced **within 0.7%**; two of six omics match exactly.
- Discrepancies live only in **tunable, data-dependent filters** — not in the pipeline logic.
- Caught a survival-label join bug that had **duplicated 6,197 samples**.
- Demographics (age) are dropped by the bottleneck — they should be **first-class model inputs**.